

Design Engineering Masters Project: Late-Stage Gateway

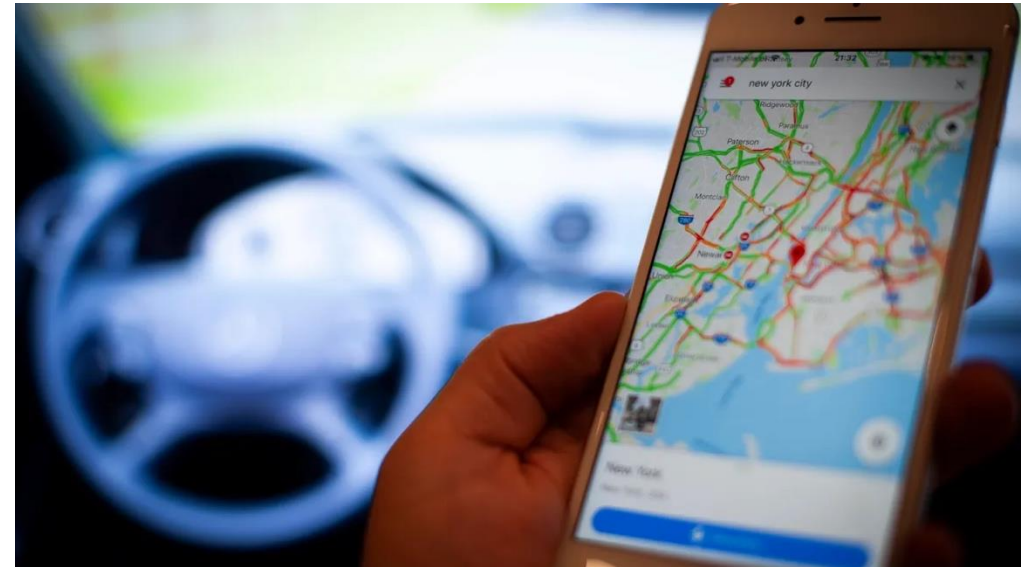
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Aaron Bendor – 02024741

Supervisor: Professor Robert Shorten

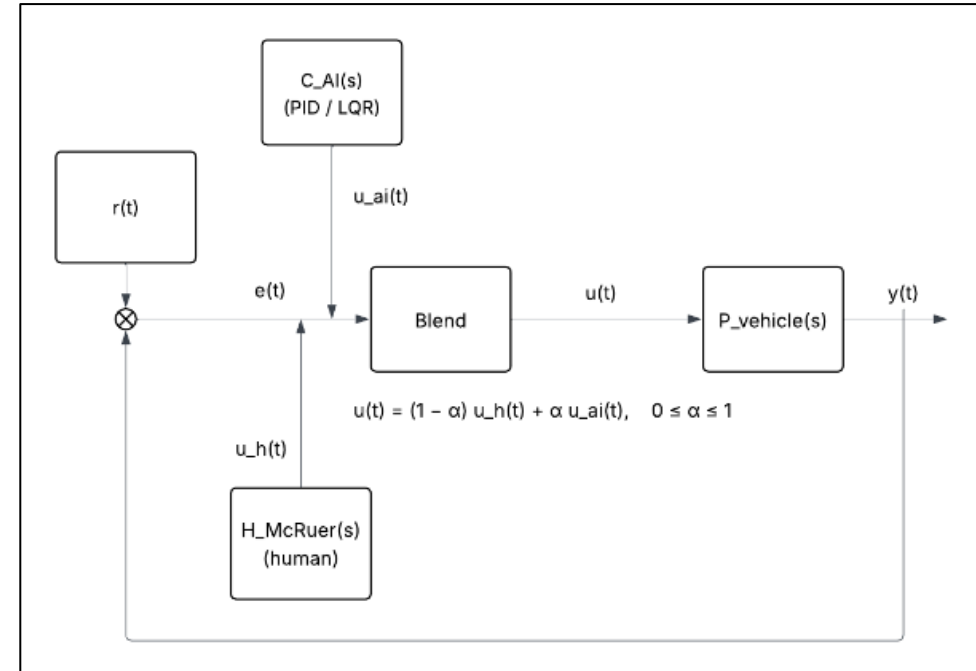
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1. Original goals

- Recover a driver's true input from vehicle signals that have already been **modified by an onboard AI assistance system** → **closed-loop system identification**
- Assess whether the method **can detect driver degradation** such as increased delay **even when AI assistance is masking** those behavioural signatures



Pivot away from closed-loop identification
due to technical feasibility

2. Updated research question

- **Can we recover a driver's intrinsic route-choice Markov chain from partially observed, externally influenced data?**
- **Motivation:**
 - Observed route choices may not reflect intrinsic behaviour.
 - Congestion increases the likelihood of external routing influence.
 - Filtering out potentially corrupted observations may improve of baseline driver-behaviour models.



Building on Morimura et al., 2013 [1]

3. Updated methodology

- I. Use an **inverse Markov-chain framework** (Morimura et al., 2013) to estimate missing transition probabilities across the network from limited synthetic traffic behaviour.
- II. Treat **periods of congestion as higher-risk periods for external route guidance** and compare observed junction decisions against the baseline model.
- III. **Filter out congestion-time deviations** that are likely to be externally influenced, so the final model better reflects the driver's **intrinsic route-choice behaviour**.



SUMO will be used to simulate the road network
[3]

4. Success metrics



Transition error: how close the recovered chain is to ground truth



Stationary error: how well long-run occupancy is recovered



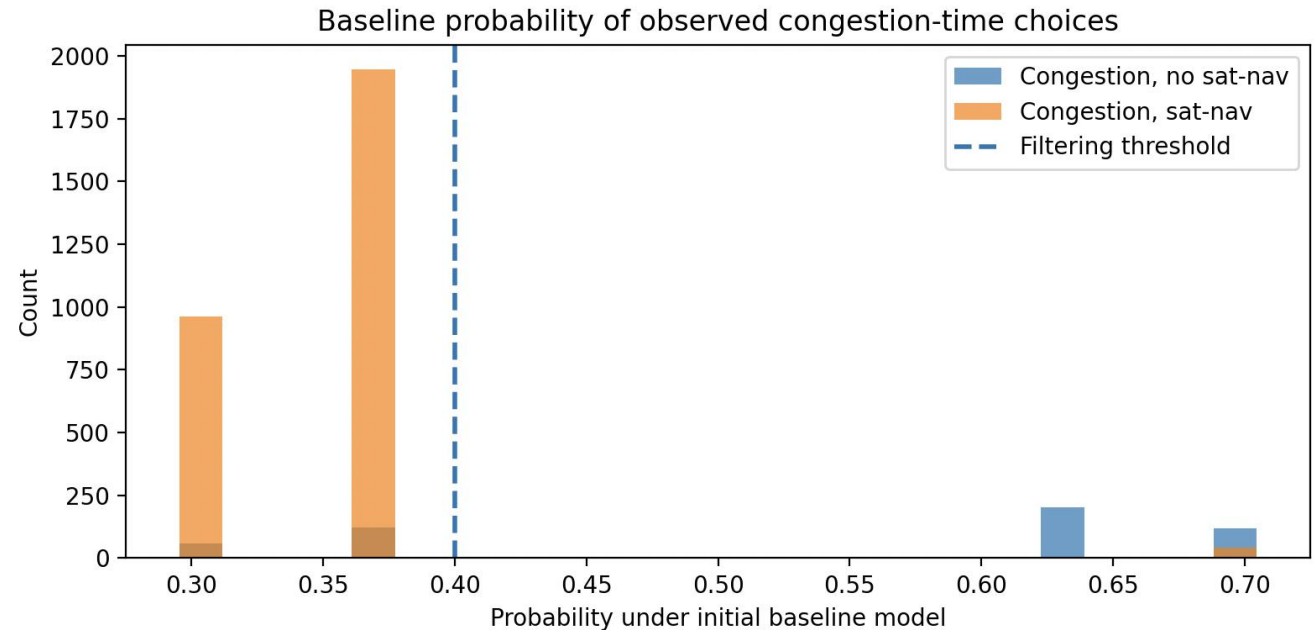
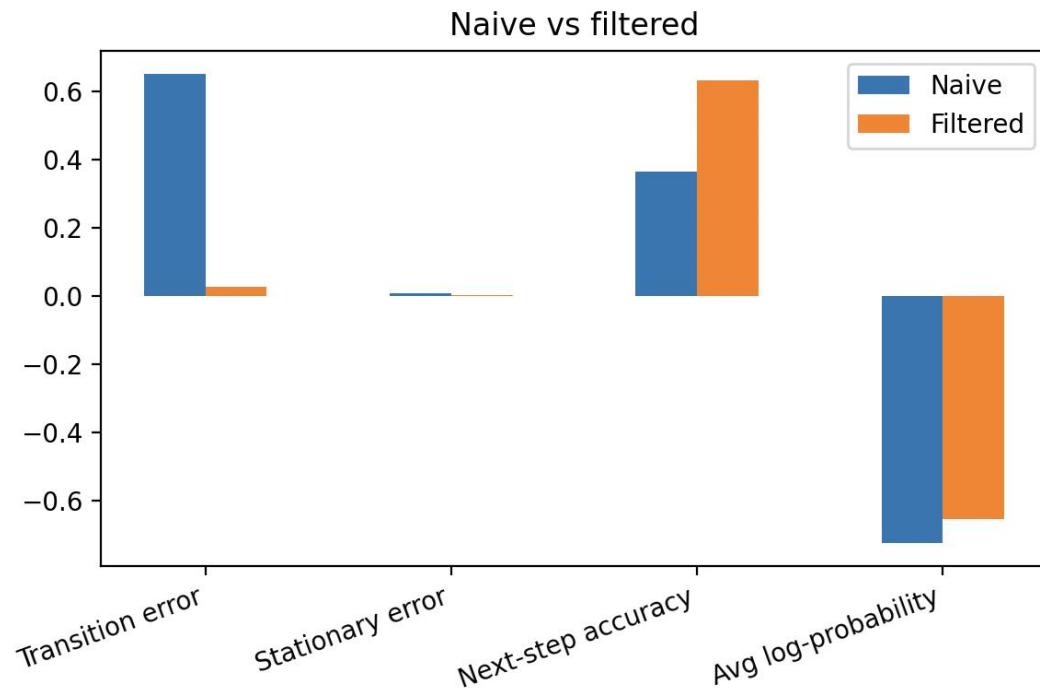
Junction prediction accuracy: how well next choices are predicted



Filtering benefit: improvement versus the unfiltered baseline

5. Current status

- Toy pipeline working → need to add noise and observe fewer states



Filtering improves recovery in the toy setting, but the current setup is still too favourable.

6. Plan for next term

- ☐ Stress-test the toy pipeline with fewer observed states, noisier congestion labels, and weaker sat-nav effects
- ☐ Implement the optimizer from Morimura et al., 2013
- ☐ Implement the road network in SUMO (using synthetic data)
- ☐ Search for real road observation data

Bibliography

1. Morimura, T., Osogami, T. and Ide, T. (2013) 'Solving inverse problem of Markov chain with partial observations', Advances in Neural Information Processing Systems, 26.
2. Crisostomi, E., Kirkland, S. and Shorten, R. (2011) 'A Google-like model of road network dynamics and its application to regulation and control', International Journal of Control, 84(3), pp. 633–651.
3. "Microscopic Traffic Simulation using SUMO"; Pablo Alvarez Lopez, Michael Behrisch, Laura Bieker-Walz, Jakob Erdmann, Yun-Pang Flötteröd, Robert Hilbrich, Leonhard Lücken, Johannes Rummel, Peter Wagner, and Evamarie Wießner. IEEE Intelligent Transportation Systems Conference (ITSC), 2018.